



Master of Information and Cybersecurity (MICS)

CYBER 295 Capstone (Section 1)

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Website Instructions: <https://www.ischool.berkeley.edu/intranet/students/mics/capstone>

Course Description

Welcome to the Master of Information and Cybersecurity (MICS) Capstone! The Capstone course is a culmination of all of your hard work in getting through the program. It provides an opportunity to dive deep into a problem and show off your skills.

It is also a rather daunting task of putting together a project from scratch with a group of your peers and presenting it to the world in just 14 short weeks. Learning to choose a good cybersecurity problem, scope it appropriately, and churn out the work for a public showcase in just 14 weeks is not easy. The goal of this course is to provide you the support and learning opportunities to help you get it done and generate a final project that you and your teammates can be proud of.

In order to fulfill this goal, the Capstone course is a mix of structure and flexibility. There is a set of fixed milestones and deadlines that are graded. There are also sections of the course that are subject to change, that we will adjust and plan for a few weeks out to accommodate pressing needs and areas where we see the class needs help and attention. In addition, given that this is an unprecedented time of difficulty caused by the global pandemic, we will balance flexibility of the learning outcomes being sought with support that is needed.

Consequently, the syllabus for this Capstone course is a living syllabus and will be updated throughout the course and adapted to the needs of the class. It is your responsibility to regularly check the syllabus for updates though we will actively cover the syllabus during each live session class when specific updates or changes are made.

The MICS Capstone course is an integrative learning experience in which students use the full range of cybersecurity skills, techniques, and concepts gained throughout the program to design and execute a semester-long team project. Student teams propose a complex cybersecurity project to which they can apply multi-faceted research, analysis, and problem-solving. Projects typically focus on real-world issues, applications, and solutions (or prototypes) with the aim of preparing students for practical and long-term success as cybersecurity professionals. For example, project topics may

include analysis, measurement, design and innovation, testing, attack, or research in technical, management, or policy domains with a focus on application of cybersecurity solutions. Students are evaluated based on the quality of their project deliverables as well as their ability to collaboratively develop, communicate, and manage their project work in both written and oral form as a team.

Throughout the course, students communicate with classmates and instructors about their work, receive and provide feedback in informal group discussions and formal class presentations, and ultimately deliver a web-based final presentation and final deliverable. The course also includes relevant, optional readings and material, case studies and discussions, and real-world examples and perspectives from cybersecurity experts and industry practitioners.

Prerequisite(s)

Students must be in their final semester of the MICS program, with exceptions made by the I School MICS program administration.

Course Goal

This course represents the Capstone experience in the Master of Information and Cybersecurity (MICS) program. It is designed for students to (a) demonstrate that they know “how to think like a cybersecurity professional”, and to (b) exercise the skills and tools they have learned throughout the program.

Course Objectives

By the completion of this course, students will be able to demonstrate:

1. Practical Application: Translate and apply classroom learning to real-world scenarios.
2. MICS Program Learning: Synthesize and integrate skills and knowledge learned throughout the MICS program.
3. Team Collaboration and Leadership: Actively participate, collaborate, and lead as part of a team toward completion of defined project goals.
4. Product and Project Management Methodology: Select and apply appropriate product and project management tools, techniques, and practices throughout the project lifecycle, to execute project milestones within schedule and scope.
5. Oral Presentation: Construct and deliver persuasive oral arguments and presentations in both formal and informal settings to a variety of audiences in various contexts (e.g., project ideas, presentation assignments, and project showcase).
6. Deliverable Design: Plan, design, and implement a complex and unique semester-long project deliverable and associated materials to address a high-impact cybersecurity problem.

Course Suggested Reading (Optional):

[Reaching the Chasm: How to Drive Your Early-Stage Start-Up to Scale](#) (Edward Amoroso)

Guidance on Project Selection

- Students are free to choose their own projects subject to instructor approval and must do so by the second week at the latest (first week is ideal)
- Come prepared with some project topic ideas for the first week. Use the course-specific Slack Canvas for discussing ideas with classmates and instructors.
- Students self-select teams by week 1. Instructors will facilitate and make team selections for those who choose to not self-organize or are unable to.
- All project ideas must be approved by instructors by week 2
- Established, self-selected teams will be respected. Edge case scenarios will be handled on a case-by-case basis, e.g., where a student indicates in week 1 that they want to switch to a new team or different project topic, or where class dynamics are severely impacted.

Course Requirements and Grading

Successful completion of this course is dependent upon a number of factors—attendance and active participation, assignments, team collaboration, and project deliverables. Each student should expect to contribute at least 10 hours per week of individual and group work to complete the project outside of weekly live session classes.

Students will form project teams and each team will be responsible for the assignments, deliverables, and presentations listed below. All team members are expected to contribute to the team deliverables and each team member is expected to present a portion of each team presentation or update.

Task or Assignment	Deadline	Point Allocation
Attendance and Participation		30%
Active Participation	Each Week	10%
Team Presentations and Updates (Every Person Speaks)	Each Week	10%
Team Peer Evaluation	Week 14	10%
Scoping and Planning		30%
1-2 Page Project Overview	Week 3	10%
Milestone Report	Week 4	5%
Solution Design & Architecture Presentation	Week 5	10%
5-Point Presentation	Week 9	5%
Implementation		40%
Demo/Proof of Concept	Week 8	10%
Final Project Showcase Site and Deliverable	Week 13	15%
Final Presentation	Week 14	15%

Key Dates

- Academic Calendar: <https://www.ischool.berkeley.edu/intranet/students/mics/calendar>
- **[Hard Requirement]** Joint Section Final Presentation Showcase: August 5, 2026 from 4:00 - 6:30 PM Pacific Time

Course Journey



W295 Capstone
Course Schedule Weeks 1 Through 14 (Subject to Change)

Week	Date	Key Activities	Deliverables
Week 1 - Ideation: Course Introduction and Initial Project Topics			
1	May 6, 2026 (Ryan)	This week will focus on the course introduction and initial project ideas: Assessing impact (being clear on who cares and why), converging project ideas (distilling to essence), focusing on a problem (not getting bogged down by technical solutions), and setting realistic and not overly ambitious scope (“Yes! And...”) - Prerequisite material to review prior to class: Primer for Convergent Thinking Versus Divergent Thinking TEDx Talk for Creative Thinking (Includes Elements of 5W Attributes and Convergent/Divergent Thinking) - Live Session: - Instructor presentation on Capstone overview - Syllabus alignment and expectations around presentations, deliverables, and participation - Team formation and project idea discussions	Due End of Live Session 1: - Project Topics Selected - Teams Formed
Week 2 - Pick Idea: Finalize Project Topics			
2	May 13, 2026 (Sekhar)	This week will focus on finalizing teams and project topics while presenting an initial scope. - Prerequisite material to review prior to class: Why minimum viable product isn't just for startups 13 Best Minimum Viable Product Examples (+8 Strategies) (Note: Ignore the emphasis on startup and vendor/agency aspects as that's not the focus of the course, just the method and concepts, unless you are personally interested) Theory of Constraints - Live Session: - Guest speaker on problem finding and solutions in cybersecurity - Instructor presentation on product management and project ideation	Due Before Live Session 2: Team Presentation (5W1H Submission Assignment Only)
Week 3 - Define MVP: Scope Refinement			
3	May 20, 2026 (Ryan)	This week will focus on changes and refinements to the initial scope, risks, obstacles encountered, and logistics for moving forward. Begin defining a minimum viable product (MVP). Prerequisite material to review prior to class:	Due Before Live Session 3: 1-2 Page Project Overview

		○ Personas, scenarios, user stories and storyboards: what's the difference? ● Live Session: ○ 10 - 15 minute presentations per team on revised problem statements and benefits of solution. Include use of frameworks such as Golden Circle, SWOT, and competitive analysis with use cases that have high level prioritization ○ Be prepared for Q&A from instructors and students to follow presentations ○ Breakout sessions further refine your project scopes	● Instructor Confirmation of Project
Week 4 - Define MVP: Project Planning			
4	May 27, 2026 (Ryan)	This week will present team Milestone Reports. ● Prerequisite material to review prior to class: ○ What is Work Breakdown Structure? ○ What is iteration? ○ Continuous Delivery Video ● Live Session ○ 10 - 15 minute presentations per team on project management execution plan, internal goals, obstacles and accomplishments, and integrated delivery method with use cases and personas defined ○ Be prepared for Q&A from instructors and students to follow presentations ○ Instructor presentation on solution architecture	Due Before Live Session 4: ● Milestone Report
Week 5 - Architecture: Solution Design			
5	Jun. 3, 2026 (Sekhar)	This week will focus on the transition from project planning to solution design. ● Prerequisite material to review prior to class: ○ Principles for the Agile Architect ● Live Session ○ 10 - 15 minute presentations per team on solution design: High level architecture and functional design with associated sequence diagrams, workflows, etc. (be sure to map back to use cases and personas previously defined) ○ Be prepared for Q&A from instructors and students to follow presentations ○ Breakout sessions to discuss projects with a focus on solution design components	Due Before Live Session 5: ● Solution Design & Architecture Presentation
Week 6 - Secure by Design: Threat Modeling			
6	Jun. 10, 2026 (Sekhar)	This week we will explore a specific security topic focused on threat modeling. ● Prerequisite material to review prior to class: ○ Security by Design Principles according to OWASP ● Live Session	

		○ A guest speaker from industry will present on threat modeling and related topics to security by design ○ Breakout sessions for teams to work on threat modeling related to their projects	
Week 7 - Secure by Design: Mid-Point (a.k.a. 5-Point) Presentation			
7	Jun. 17, 2026 (Ryan)	This week will be focusing on 5-Point presentations that will include updates on use cases, architecture, and solution implementation with the incorporation of secure by design principles and threat modeling. ● Prerequisite material to review prior to class: ○ Details will be emailed (bCourses Announcement) ● Live Session ○ 10 - 15 minute presentations per team to present and discuss ○ Be prepared for Q&A from instructors and students to follow presentations ○ Breakout sessions to discuss projects with a focus on approach and schedule (realism check, pending time)	Due Before Live Session 7: ● 5-Point Presentation
Week 8 - Demo: Mid-Point Demo			
8	Jun. 24, 2026 (Sekhar)	This week will be a demonstration of team project demos or proofs of concept to the class. ● Prerequisite material to review prior to class: ○ How to Present an Easy-to-Follow Tech Demo ● Live Session ○ 20 minute presentations including a demo or proof of concept ○ Be prepared for Q&A from instructors and students to follow presentations ○ Breakout sessions to focus on demo feedback and team execution next steps (if time remaining)	Due Before Live Session 8: ● Demo/Proof of Concept
Week 9 - Demo: Solution Bias and Privacy			
9	Jul. 1, 2026 (Ryan)	This week we will continue threat modeling integration and explore key topics to reflect interdisciplinary aspects of your solutions ranging from emerging bias to privacy. ● Prerequisite material to review prior to class: ○ The Belmont Report ○ Digital Global Warming ○ Systemic Racism Is a Cybersecurity Threat ○ Great Promise but Potential for Peril ● Live Session ○ Lecturer-facilitated discussion ○ Breakout sessions for teams to perform related exercises ○ As a class, 10 - 15 minute team presentations on key learnings from breakout sessions	
Week 10 - Pitch: Retrospective on Competitiveness			

10	Jul. 8, 2026 (Sekhar)	This week will use retrospectives and related project delivery concepts to consider how originality, impact, and completeness are integrated into your presentations. ● Prerequisite material to review prior to class: ○ Competitive Analysis: How To Conduct A Comprehensive Competitive Analysis ● Live Session ○ Breakout sessions focused on concepts of originality, impact, and completeness ○ Class plenary brief backs based on breakout session work	
Week 11 - Pitch: Impactful Presentation			
11	Jul. 15, 2026 (Ryan)	This week will provide learning opportunities on the creation of impactful presentations. ● Prerequisite material to review prior to class: ○ Best Practices For Virtual Presentations ● Live Session ○ A guest speaker from industry will present on impactful presentations ○ 2 minute pitch per project team with feedback	Due Before Live Session 11: ● 2 Minute Pitches
Week 12 - Final Presentation: Demo Updates and Dry Run Presentations			
12	Jul. 22, 2026 (Sekhar)	This week will focus on teams presenting final, or near-final, solution demos as an initial dry run of final presentations. Instructors will offer feedback and guidance.	Due Before Live Session 12: ● Final Project Deliverables Webpage
Week 13 - Final Presentation: Final Web Deliverable and Presentation			
13	Jul. 29, 2026 (Ryan)	This week will be the last run through before the final presentation. Opportunity to refine and “fine tune” final presentations.	Due Before Live Session 13: ● Final Project Deliverables
Week 14 - Final Presentation: Showcase			
14	Aug. 5, 2026 <i>Combined Sections</i> (Student Affairs)	Final Presentation showcase with an industry expert judging panel.	Due Before Live Session 14: ● Final Presentation

Attendance, Participation, and Collaboration

Student preparation, attendance, and participation are required in live session classes. This course is framed by the notion that all students are not only learners but educators as well. The depth and richness of this course and the forthcoming discussions, therefore, can only be achieved if students regularly attend class, are prepared in advance, and work collaboratively with their team members. Active participation of each student is expected every week and will be taken into account with grading. Speaking during live session group discussions, asking

questions, and posting relevant insights into the live session chat is considered active participation. Students may also supplement active participation through the use of the course Slack channel; however, this may not be used as a substitute for attending and actively participating in live session classes.

Where students are unable to attend a live session class, instructors and team members must be notified before the start of the live session. Participation and presentation grades will also consist of team peer evaluations.

Expectations for Online Interaction

All students will be held to the highest standards for language and content in all interactions. Abusive and derogatory language or unacceptable content will not be tolerated. For expectations regarding online interaction, please see "The Core Rules of Netiquette", from the book *Netiquette* (ISBN: 0-9637025-1-3) by Virginia Shea at: <http://www.albion.com/netiquette/corerules.html>

Academic Integrity

All students—undergraduate, graduate, professional full time, part time, law, etc.—must be familiar with and abide by the provisions of the “Student Code of Conduct” including those provisions relating to Academic Misconduct. All forms of academic misconduct, including cheating, fabrication, plagiarism or facilitating academic dishonesty will not be tolerated. The full text of the UC Berkeley Honor Code is available at: <https://teaching.berkeley.edu/berkeley-honor-code> and the Student Code of Conduct is available at: https://sa.berkeley.edu/student-code-of-conduct#102.01_Academic_Misconduct

AI Usage

Students are encouraged to explore and utilize AI applications (including generative AI) and tools as part of their learning process. Students are also encouraged to make suggestions about use of AI tools in class presentations, discussions, and communications (e.g., course Slack). While the use of AI is permitted, **students remain fully responsible for the work produced and submitted**. This includes, but are not limited to:

- **Accuracy and Critical Evaluation:** AI-generated content can be inaccurate, incomplete, or biased. It is the students' responsibility to critically evaluate, fact-check, and verify all information obtained from AI tools. Submitting work with unverified or erroneous AI-generated information will impact grades.
- **Original Thought and Contribution:** AI should be used as a tool to support student learning and work, not to replace critical thinking, analysis, and original contributions. The core ideas, arguments, and insights in student work must be their own.
- **Understanding and Mastery:** Students are expected to understand the material submitted during the course. Be prepared to discuss, elaborate on, and defend any portion of work, regardless of whether AI was used in its preparation.
- **Avoiding Plagiarism:** Representing AI-generated work as entirely your own without proper acknowledgement is a form of academic dishonesty. Proper citation guidelines must be followed.

Citation Guidance:

- **In-text Citation**
 - Include an in-text citation when incorporating content from a generative AI tool into your academic paper. For example:
 - "UC Berkeley is a top-tier School of Information due to its academic excellence, innovative curriculum, strong alumni network, prime location, diverse community, emphasis on leadership and ethics, and abundant experiential learning opportunities (OpenAI, 2025)."
- **Reference List Entry:**

- In your reference list, provide the full details of the source. Since ChatGPT is an online tool, provide the publication date (if available), title, and URL:
- Source: Gemini. (2025). Gemini (May 1 2.5 Pro version) [Large language model]. <https://gemini.google.com>

Grading Scale

The following grading ranges (in %) will be applied for course grade submissions:

A+ = 100+ and Instructor Discretion

A= 99.9 - 94.0

A- = 93.9 - 90.0

B+ = 89.9 - 87.0

B = 86.9 - 83.0

B- = 82.9 - 80.0

C+ = 79.9 - 77.0

C = 76.9 - 73.0

C- = 72.9 - 70.0

D+ = 69.9 - 67.0

D = 66.9 - 63.0

D- = 62.9 - 60.0

F = 59.9, and below

Disability Services & Accommodations

The I School recognizes disability in the context of diversity, and the Disabled Students' Program (DSP) equips students with appropriate accommodations and services to remove barriers to educational access. Students seeking accommodations in this class are responsible for completing the DSP application process to obtain an accommodation letter. (510) 642-0518, <https://dsp.berkeley.edu>

Emergency Preparedness

To Report a Campus Emergency or Suspicious Activity: Call the University Police Department at 510-642-3333. If the line is unavailable or you are calling from another University location or off campus, dial 911. Detailed resources on can be found at: <https://ucpd.berkeley.edu/>

Accommodation of Religious Creed

In compliance with Education code, Section 92640(a), it is the official policy of the University of California at Berkeley to permit any student to undergo a test or examination, without penalty, at a time when that activity would not violate the student's religious creed, unless administering the examination at an alternative time would impose an undue hardship which could not reasonably have been avoided. Requests to accommodate a student's religious creed by scheduling tests or examinations at alternative times shall be submitted directly to the faculty member responsible for administering the examination. Reasonable common sense, judgment and the pursuit of mutual goodwill should result in the positive resolution of scheduling conflicts. The regular campus appeals process applies if a mutually satisfactory arrangement cannot be achieved.

<https://sa.berkeley.edu/uga/religion>

Equity and Inclusion

Integrating a diverse set of experiences is important for a more comprehensive understanding of cybersecurity. We will make an effort to read papers and hear from a diverse group of practitioners, still, limits exist on this diversity in the field of cybersecurity. I acknowledge that it is possible that there may be both overt and

covert biases in the material due to the lens with which it was created. I would like to nurture a learning environment that supports a diversity of thoughts, perspectives and experiences, and honors your identities (including race, gender, class, sexuality, religion, ability, veteran status, etc.) in the spirit of the UC Berkeley Principles of Community <https://diversity.berkeley.edu/principles-community>

To help accomplish this, please contact me or submit anonymous feedback through I School channels if you have any suggestions to improve the quality of the course. If something was said in class (by anyone) or you experience anything that makes you feel uncomfortable, please talk to me about it. If you feel like your performance in the class is being impacted by experiences outside of class, please don't hesitate to come and talk with me. I want to be a resource for you. Also, anonymous feedback is always an option, and may lead to me to make a general announcement to the class, if necessary, to address your concerns. As a participant in teamwork and course discussions, you should also strive to honor the diversity of your classmates.

If you prefer to speak with someone outside of the course UC Berkeley [Office for Graduate Diversity](#) is an excellent resource. Also, see the following: <https://www.ischool.berkeley.edu/about/community>.